International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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Student identification

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Platform used	Windows 11 Home (Host Environment) with PowerShell & DiskPart
Submission file	CIP-B101_Lab1B_Kenneth_Ukata.pdf

1. Executive Summary & Lab Purpose

The objective of this laboratory exercise is to inspect computer system fundamentals, hardware specifications, and storage partition structures using non-destructive, read-only administrative tools. In digital forensics, Analysts must be capable of extracting critical system metadata and verifying disk partition geometries without modifying the underlying evidence sources.

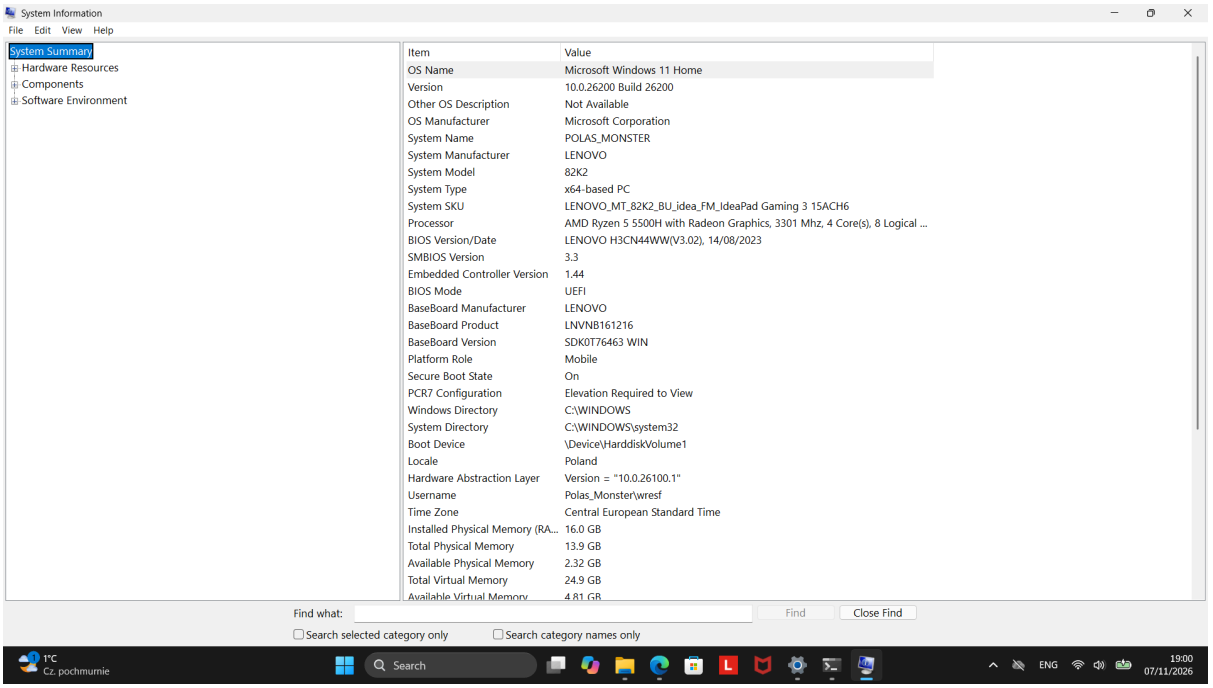
All commands executed during this session were strictly observational, complying fully with evidence preservation standards.

2. Part A – System Identification and Evidence Layer Mapping

A1. Host System Specifications

System hardware and operating system parameters were gathered using the built-in Windows System Information utility (`msinfo32`).

- **Operating System:** Microsoft Windows 11 Home (Version 10.0.26200 Build 26200)
- **System Manufacturer & Model:** LENOVO (Model: 82K2 – IdeaPad Gaming 3 15ACH6)
- **Processor:** AMD Ryzen 5 5500H with Radeon Graphics (3.30 GHz, 6 Cores, 12 Threads)
- **System Type:** x64-based PC
- **BIOS Mode:** UEFI (Secure Boot State: On)
- **Installed Physical Memory (RAM):** 16.0 GB (13.9 GB Total Usable)



A2. Storage Controller & Logical Volume Identification

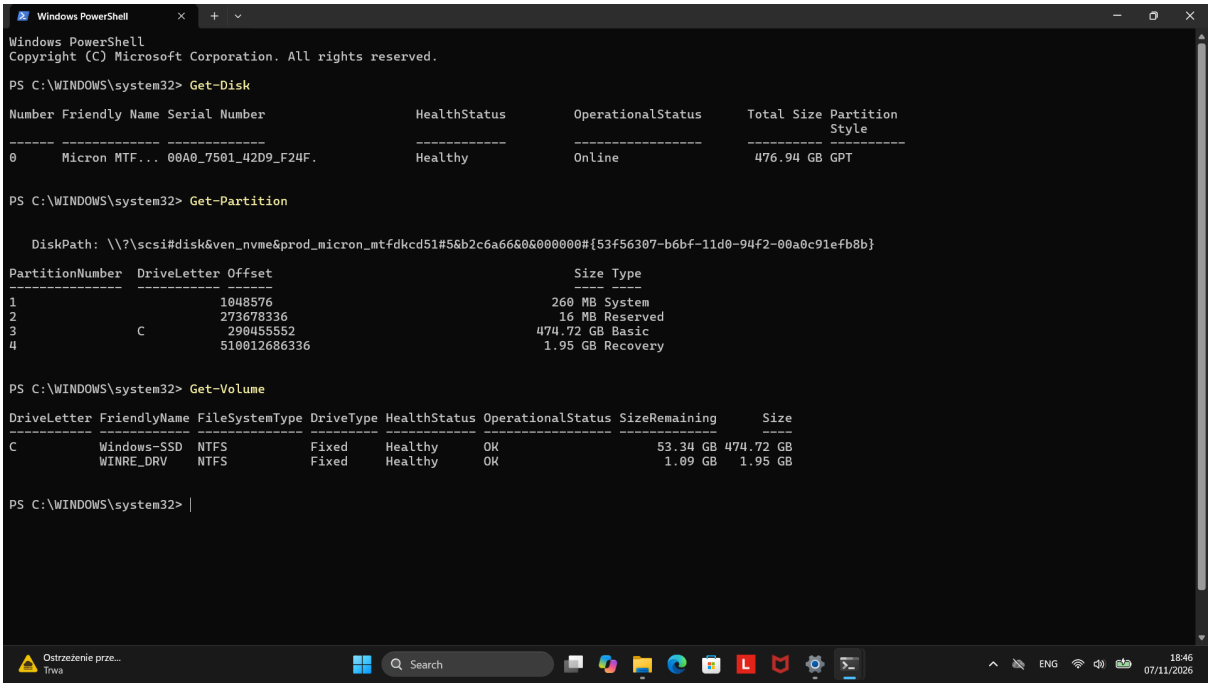
Storage hardware and logical drive volumes were enumerated via read-only PowerShell storage cmdlets (`Get-Disk` and `Get-Volume`).

Physical Disk Specifications:

- **Disk Number:** Disk 0
- **Model: Micron** MTFDKCD512TFK (NVMe Solid State Drive)
- **Total Capacity:** 476.94 GB
- **Partition Style:** GPT (GUID Partition Table)
- **Health Status:** Healthy (Online)

Logical Volumes Identified:

1. **C: Drive (Windows-SSD):** File System: **NTFS** | Size: 474.72 GB | Free Space: 53.34 GB
2. **WINRE_DRV (Recovery):** File System: **NTFS** | Size: 1.95 GB | Free Space: 1.09 GB



A3. Evidence Source Layer Mapping Table

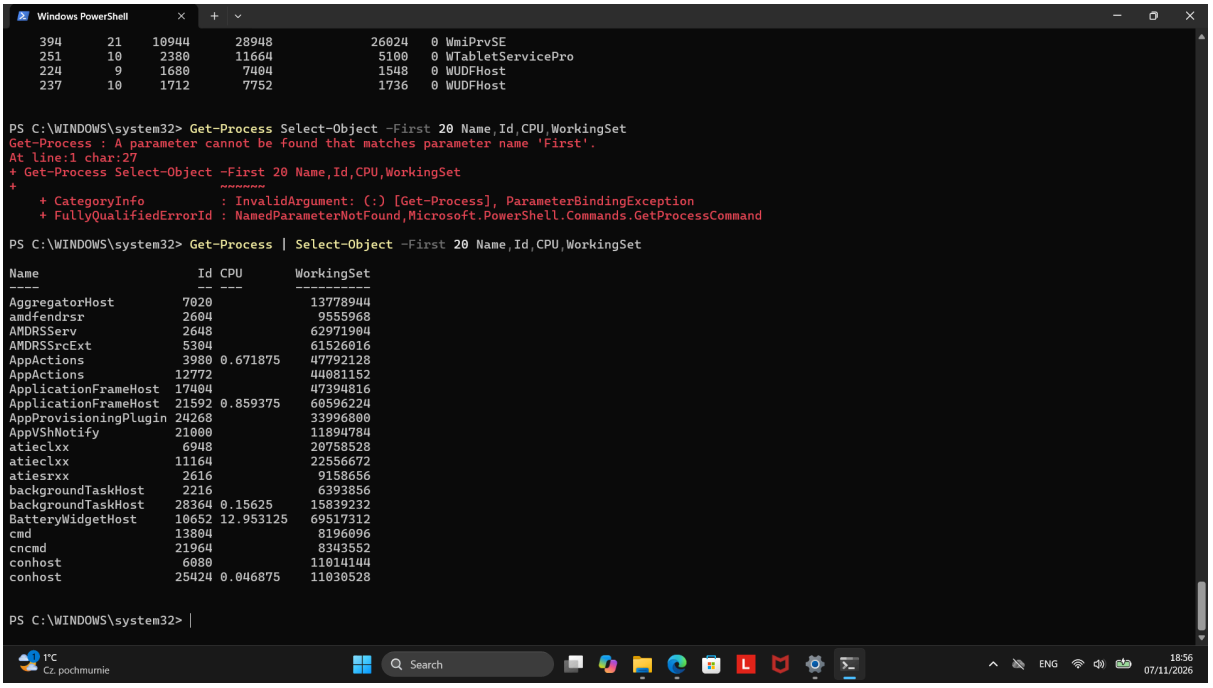
Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Recent documents, browser search history, shell activity	User profile folders (**`C:\Users\`**), web cache, bash history	Connects specific user actions directly to system events.
Application layer	Office document metadata, browser cookies, SQLite databases	**`AppData\Roaming`**, application logs	Proves which specific software created or altered evidence.
OS layer	System Event Logs, registry hives, account login history	**`C:\Windows\System32\winevt\Logs`**, Windows Registry	Tracks system-level actions, user logins, and installed software.
Disk layer	Deleted files, unallocated clusters, file system journals	Master File Table (**`$MFT`**), **`$LogFile`**, unallocated space	Enables file recovery and chronological timeline analysis.
Memory/process layer	Active malware processes, unencrypted memory data, network connections	RAM dump, live process list (**`Get-Process`**)	Captures volatile evidence that disappears when the PC shuts down.

Why Evidence Exists Across Multiple Layers: Digital actions trigger sequential events through the technology stack. For instance, downloading a file starts at the User Layer (browser click), moves to the Application Layer (browser cache writing), interacts with the OS Layer (security logging), and writes data to the Disk Layer (file block allocation).

3. Part B – Program, Process, and Thread Investigation

B1. Active Process Observation

System processes were observed in PowerShell using `Get-Process`.



Selected Process Evaluation:

1. **`cmd.exe` (PID: 13804):**
   - *Purpose:* Windows Command Prompt interpreter. Used by system administrators or users to run command-line tools.
2. **`conhost.exe` (PID: 6080):**
   - *Purpose:* Console Window Host process. Provides the interface window for command-line applications like `cmd.exe` or PowerShell.
3. **`AMDRSServ.exe` (PID: 2648):**
   - *Purpose:* AMD Radeon Software background service. Manages graphics card display settings and hardware monitoring.

B2. Theoretical Concepts & Forensic Relevance

- **Program:** A static executable file saved on non-volatile disk storage (e.g., `C:\Windows\System32\cmd.exe`). It contains instructions but is not executing.
- **Process:** An active instance of a program loaded into volatile memory (RAM). It possesses an isolated memory space, environment variables, and a unique Process ID (PID).
- **Thread:** The smallest executable unit of code within a process. A single process can spawn multiple threads that execute concurrently to perform tasks.
- **Forensic Relevance:** Inspecting live processes during triage helps investigators spot unauthorized background tasks, memory-only malware, privilege escalation attempts, or suspicious outbound network connections before the PC is powered off.

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4. Part C – Disk Sector and Disk Size Calculations

C1. Reproducing Slide Calculation

Using CHS (Cylinder-Head-Sector) geometry parameters from the lecture materials:

Parameters: 400 sectors/track, 12 heads, 17,000 cylinders, 512 bytes per sector.

1. **Total Sectors:**

$$400 \times 12 \times 17,000 = 81,600,000 sectors$$

2. **Total Bytes:**

$$81,600,000 \times 512 = 41,779,200,000 bytes$$

3. **Kilobytes (KB):**

$$41,779,200,000/1024 = 40,800,000 KB$$

4. **Megabytes (MB):**

$$40,800,000/1024 = 39,843.75 MB$$

5. **Gigabytes (GB - Binary / GiB):**

$$39,843.75/1024 = 38.91 GB$$

5. Part D – Safe Partition Table Inspection

The partition layout of **Disk 0** was inspected using the `diskpart` utility in a read-only session.

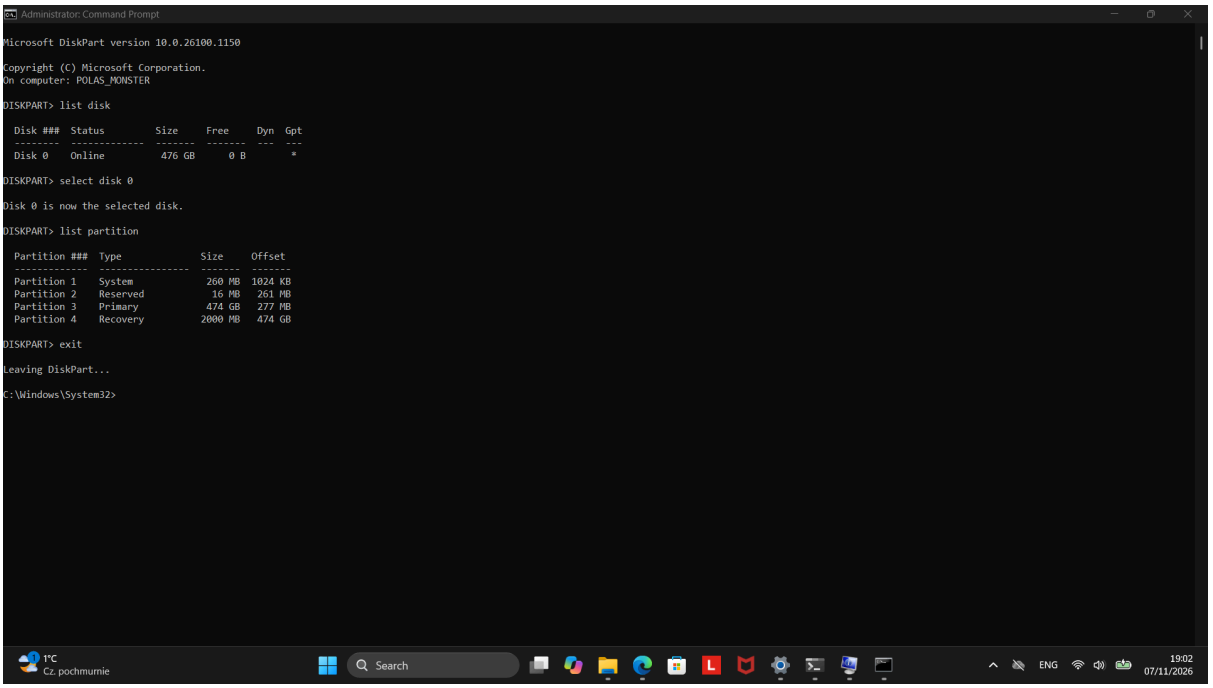
```
diskpart

list disk

select disk 0

list partition
```

```
exit
```



Partition Table Summary:

- **Disk Type:** GPT (GUID Partition Table)
- **Partition 1 (System):** 260 MB (Offset: 1024 KB)
- **Partition 2 (Reserved):** 16 MB (Offset: 261 MB)
- **Partition 3 (Primary):** 474 GB (Offset: 277 MB)
- **Partition 4 (Recovery):** 2000 MB / 1.95 GB (Offset: 474 GB)

6. Part E – MBR, GPT, and Boot Process Explanation

E1. Computer Boot Sequence Flow

1. **Power-On & POST:** Power is supplied to hardware components. Firmware runs the Power-On Self-Test (POST) to verify memory and CPU integrity.
2. **BIOS/UEFI Firmware Initialization:** Hardware initialization occurs. UEFI firmware checks firmware settings stored in NVRAM and selects the default boot device.
3. **EFI System Partition / Boot Record Access:** On UEFI systems, the firmware reads the EFI System Partition (ESP) to locate boot loaders listed in boot entries.
4. **Boot Loader / OS Manager:** The Windows Boot Manager (`bootmgr.efi`) initializes, presenting multi-boot menus if configured, and loads the Windows OS kernel (`ntoskrnl.exe`).
5. **Kernel Execution & OS Control:** The OS kernel assumes control of the system, initializes drivers, starts background services, and presents the user login prompt.

- **Local System Boot Mode: UEFI Mode** (verified via `msinfo32`).

E2. MBR vs. GPT Comparison Matrix

Feature	MBR (Master Boot Record)	GPT (GUID Partition Table)
Introduced / Era	Legacy (1980s standard)	Modern (UEFI standard)
Maximum Disk Size	~2.0 TB limit	~9.4 Zettabytes (ZB) limit
Partition Limit	Max 4 Primary partitions (or 3 Primary + 1 Extended)	Up to 128 Primary partitions in Windows
Redundancy & Recovery	Single partition table stored in Sector 0 (single point of failure)	Primary header at start of disk; secondary backup header at end of disk

7. Part F – File System Identification and Interpretation

F1. Identified File Systems

- **C: Drive (Windows-SSD): NTFS** (New Technology File System)
- **WINRE_DRV (Recovery Drive): NTFS**

F2. Forensic Importance of File Systems

File systems determine how an operating system writes, structures, and indexes data on disk sectors. For digital investigators, file systems are critical because:

- **Metadata Tracking:** Modern file systems like NTFS maintain rich structural metadata within the Master File Table (`$MFT`), including file owner IDs, access permissions, and MACB (Modified, Accessed, Created, MFT-Changed) timestamps.
- **File Recovery:** Knowing the file system structure allows examiners to recover deleted files from unallocated clusters using file carving techniques.
- **Journaling Integrity:** File system journals (e.g., NTFS `$LogFile` or `$UsnJrnl`) track pending disk writes, providing evidence of user file modifications even if files were later deleted.

8. Part G – Final Practical Assignment (Partitions 1 & 2 Size Calculations)

Partition Analysis & Sector Size Verification

Task 1: Partition 1 Analysis (EFI System Partition)

- **Partition Type:** System (Boot loader storage)
- **Reported Size:** 260 MB
- **Sector Calculation Step-by-Step:**

Converting 260 MB into bytes:

- $260MB \times 1024KB/MB \times 1024Bytes/KB = 272,629,760Bytes$

- Converting Bytes to Binary Gigabytes (GiB / GB):

  - $SizeinGB = \frac{272,629,760}{1024 \times 1024 \times 1024} = 0.2539GB$

Task 2: Partition 2 Analysis (Microsoft Reserved Partition / MSR)

- **Partition Type:** Reserved
- **Reported Size:** 16 MB
- **Sector Calculation Step-by-Step:**

  Converting 16 MB into bytes:

  - $16MB \times 1024KB/MB \times 1024Bytes/KB = 16,777,216Bytes$

- Converting Bytes to Binary Gigabytes (GiB / GB):

  - $SizeinGB = \frac{16,777,216}{1024 \times 1024 \times 1024} = 0.0156GB$

Task 3: Comparison with OS-Reported Values & Discrepancy Explanation

- **Comparison:** DiskPart reports Partition 1 as 260 MB and Partition 2 as 16 MB, which matches our calculated binary values (0.254 GB and 0.016 GB).
- **Explanation of Differences:** Minor variations between reported and calculated storage sizes usually occur because storage manufacturers calculate capacities using base-10 decimal units ($1GB = 10^9 bytes$), whereas operating systems calculate capacity using base-2 binary units ($1GiB = 1024^3 bytes = 1,073,741,824bytes$). Additionally, file system formatting reserves a small percentage of sectors for metadata tables and block alignment.

9. Final Conclusion

This laboratory practical demonstrated how to perform non-destructive system identification and partition analysis. By analyzing host specs, mapping evidence across the five system layers, evaluating running processes, and manually calculating partition sizes, I gained practical experience in verifying digital storage configurations. Understanding these underlying concepts ensures that evidence collected during real-world triage remains admissible, accurate, and fully verifiable.